REFINED VALUATIVE CRITERION: UNIQUENESS OVER A DENSE OPEN IMPLIES SEPARATEDNESS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Refined valuative criterion proper. Limits, Lemma lemma-refined-valuative-criterion-proper (same chapter)
- Quasi-separatedness over locally Noetherian bases. Properties, Lemma properties-lemma-locally-Noetherian-quasi-separated

Source reference. Original lemma 0CM3: sources/stacks-chow/tags.txt, source lines 15843–15843. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01OY: sources/stacks-chow/tags.txt, source lines 2211–2211. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KV: sources/stacks-chow/tags.txt, source lines 2064–2064. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ and $h : U \to X$ be morphisms of schemes. Assume that S is locally Noetherian, that f is locally of finite type, that h is of finite type, and that $h(U)$ is dense in X. If given any commutative solid diagram*

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U & \xrightarrow{h} & X \\ \downarrow & \nearrow & & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & & \longrightarrow & & S \end{array}$$

where A is a discrete valuation ring with field of fractions K, there exists at most one dotted arrow making the diagram commute, then f is separated.

Proof. We will apply Lemma 0CM3 to the morphisms $U \to X$ and $\Delta : X \to X \times_S X$. We check the conditions. Observe that Δ is quasi-compact by Properties, Lemma 01OY (and Schemes, Lemma 01KV). Of course Δ is locally of finite type and separated (true for any diagonal morphism). Finally, suppose given a commutative solid diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U & \xrightarrow{h} & X \\ \downarrow & & & \nearrow & \downarrow{\scriptstyle \Delta} \\ \mathrm{Spec}(A) & & \xrightarrow{(a,b)} & & X \times_S X \end{array}$$

where A is a discrete valuation ring with field of fractions K. Then a and b give two dotted arrows in the diagram of the lemma and have to be equal. Hence as dotted arrow we can use $a = b$ which gives existence. This finishes the proof. □

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